Sahaj Bhadja

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Summary

Junior software engineer with 2+ years supporting web applications, reports, and data workflows using Java, Python, JavaScript, React, SQL, and Spring Boot. Experienced in data validation, bug fixes, system monitoring, unit and regression testing, Agile delivery, and documentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Database Systems, Web Technologies, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python APIs, Azure SQL services, and operational data views for 500+ users, optimizing indexed schemas and report queries to reduce response latency from 1.2 seconds to 350 milliseconds.
- Investigated processing discrepancies using logs, job states, and validation results, implementing maintainable fixes and documentation that tripled throughput while improving data completeness, system monitoring, and production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Supported Flask web applications handling 10,000+ daily requests, implementing minor enhancements, authentication, data workflows, and error handling while resolving application issues affecting more than 150 users.
- Built React status interfaces and processing for 1,000+ files with relational persistence, data validation, automated tests, monitoring, and recovery, reducing turnaround 60% under concurrent production workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python data pipelines processing 100,000+ records, improving analysis accuracy 18% through completeness checks, reproducible experiments, automated validation, performance monitoring, clearly documented logic, and systematic debugging.
- Created structured analysis outputs combining map geometry and agent state, improving stability 15% while documenting data sources, reviewing reported discrepancies, and validating results across diverse simulated scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed Python services and SQL reporting integrations using AWS Glue, S3, filtering, pagination, and caching, improving response times 30% and processing efficiency 68% across connected enterprise applications.
- Maintained 20+ unit, integration, and regression suites with Jenkins, PyTest, and Mockito, accelerating discrepancy investigation, bug verification, code reviews, and reliable CI/CD releases across multiple environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed JavaScript and Node.js enhancements across eight microservices with authentication, REST APIs, WebSocket updates, and state synchronization, securely processing more than two million Azure SQL database records.
- Automated functional tests, regression checks, staged deployments, and system monitoring through Jenkins, Docker, and Azure, reducing release time 50% while improving maintenance consistency and overall production availability.

Skills

Languages & Frontend: Java, Python, JavaScript, TypeScript, SQL, React, Angular.js, Next.js, HTML, CSS, object-oriented programming, data structures, algorithms
Applications & Data: Spring Boot, Node.js, Flask, REST APIs, PostgreSQL, Azure SQL, SQL Server, MySQL, MongoDB, report queries, relational design, data validation
Testing & Support: Unit testing, integration testing, regression testing, PyTest, Mockito, debugging, discrepancy analysis, log analysis, system monitoring, bug fixes, application maintenance
Delivery & Documentation: Git, code reviews, Jenkins, GitHub Actions, CI/CD, Docker, Azure, Jira, Agile/Scrum, stand-ups, sprint planning, retrospectives, technical tasks, report documentation

Projects

Chat2SQL | Code | *Java, TypeScript, SQL, Docker, Maven*
- Built a Java and TypeScript reporting tool supporting 10+ databases with schema-aware SQL generation, query validation, metadata exploration, discrepancy handling, automated testing, and Docker-backed CI/CD deployment.

PiXMod | Live | *React, Next.js, JavaScript, MongoDB, Stripe*
- Built a responsive full-stack product using React, Next.js, MongoDB, and Stripe, implementing maintainable user workflows, API integrations, automated validation, and subscription features that increased premium conversion 18%.

HMIS | Live | *MERN, TypeScript, REST APIs, Dashboards, Sentry*
- Built a TypeScript workflow platform with operational dashboards, REST APIs, role-based access, scheduling, notifications, and Sentry monitoring, reducing administrative workload 30% through validated data and documented releases.